

A Derived Response Cutoff for Elastic Dark-Energy Lensing: $y_c = Z/2$ from Verlinde's Own Entropy Budget

Two machine-verified results pricing the elastic-medium branch of the de Sitter–Unruh program: the response cutoff is *located* at $y_c = Z/2 \approx 2.89$ (not detected), and the medium's shear sector provably cannot be linearized general relativity ($\rho_\Lambda c^2/K_{\text{eff}} = 6Z^2 \approx 201$, exact)

Carl P. Zimmerman · Briar Creek Tech · 2026-07-10

Abstract

In the de Sitter–Unruh reading of the MOND acceleration scale, $a_0 = c^2 \sqrt{\Lambda/32\pi} = cH_\Lambda/Z$ with $Z = \sqrt{32\pi/3} \approx 5.79$, the GW170817 photon–graviton bound forces any dark-matter-free lensing enhancement into the *one shared metric* as a source. One candidate realization — **Branch B** of the program — is an elastic response of the dark-energy medium to baryonic displacement, in the class of Verlinde's emergent-gravity mechanism, whose deep-regime amplitude was previously found to land at $\sqrt{Z/6} = 0.982$ of the required lensing source. This note reports two results that *price* that branch; neither is a detection and neither claims the branch. **(1) The response cutoff is derived, and located.** For the medium whose only nonlinearity is volumetric (the det-class), the constitutive law matched to the deep-MOND displacement is scale-free, so the response *must* terminate at the entropy-budget scale rather than inside the constitutive law — indeed a single-valued volumetric law provably cannot produce the required high-acceleration tail (a double-valuedness/spinodal theorem). The termination point follows from Verlinde's own budget: taking his matter entropy displacement $S_M = -2\pi M c r/\hbar$ and dark-energy entropy $S_{DE} = (r/L) A/4G\hbar$ *verbatim from the published version of record* (SciPost Phys. 2, 016 (2017), eqs. (28)–(34)), the strain–budget ratio is exactly $\varepsilon_M = 2 g_{\text{bar}}/a_{0V}$, so the crossover sits at $g_{\text{bar}} = a_{0V}/2$, i.e. $y_c \equiv g_{\text{bar}}/a_0 = Z/2 = 2.894$. The one competing $O(1)$ reading ($y_c = Z/3$, from the unabsorbed $(d-1)/(d-2)$ volume factor) is excluded by Verlinde's own eq. (41). The minimally budget-consistent throttle passes the Saturn ephemeris bound with an 18–26× margin, leaves the baryonic Tully–Fisher zero-point untouched to machine precision, and preserves the deep-regime 0.982 amplitude. **(2) The medium's shear sector cannot be linearized GR.** The hypothesis that the medium's shear waves *are* gravitational waves (which would make its shear anharmonicity negligible by the Planck-stiffness hierarchy) is refuted on the program's own normalizations: linearized GR pins only $\mu/\rho = c^2$, while the deep-MOND energy match, together with the derived compatibility lemma $J_2 = J_1^2/6$ for the radial displacement profile, caps the medium's shear modulus a factor $\rho_\Lambda c^2/K_{\text{eff}} = 6Z^2 = 201.1$ (exact) below GR stiffness — so either the 0.982 amplitude collapses to 0.119 (excluded by rotation-curve data) or the medium's shear waves propagate at $\lesssim 0.17c$ and are not gravitational waves. **The observational status is stated plainly:** the cutoff's fingerprint — a kink at y_c with a peak deviation of 0.017 dex at $y \approx 6$ in the radial-acceleration relation — is currently *indistinguishable* from the uncut modified-inertia alternative (0.5 σ with existing SPARC data; the discriminating window is bulge-dominated and the bulge mass-to-light systematic is $2.5 \times$ the signal). No data preference in either direction is claimed. This is a located, not detected, signature, with the decisive test specified: a

bulge-rich rotation-curve sample with external stellar-population mass-to-light priors at the ~ 0.005 dex level.

1. Context: why a source-side medium, and what this note does not claim

The de Sitter–Unruh program reads a_0 as set by the dark-energy density. Its modified-*inertia* realization evades the solar-system tests that exclude modified-gravity MOND, but — light being massless — under-predicts gravitational lensing on a single metric, and the GW170817 photon–graviton speed bound ($|\Delta c|/c \lesssim 10^{-15}$) excludes photon-only (disformal/bimetric) fixes by five to seven orders of magnitude, as it excluded the TeVeS class [Boran et al. 2018]. Any dark-matter-free lensing enhancement must therefore be a *source* in the one shared metric. The program thus carries a fork: **Branch A** retains modified inertia and concedes that lensing requires a genuine dark component; **Branch B** realizes the lensing source as an elastic response of the dark-energy medium itself — Verlinde’s mechanism class [Verlinde 2017] — which reproduces the required source’s radial shape, its $\sqrt{M_{\text{bar}}}$ per-galaxy scaling, and (taking the displacement scale to be the program’s horizon quantity $a_{0V} = cH_\Lambda = Za_0$) its amplitude to $\sqrt{Z/6} = 0.982$. Branch B is *not adopted here*: it replaces the modified-inertia mechanism, re-exposes the Cassini quadrupole channel, and carries posited ingredients. This note prices two of those ingredients — one is derived away, one is refuted — and characterizes the branch’s observable signature honestly.

2. Result 1: the response cutoff is necessary, and its location is $y_c = Z/2$

Necessity. Model the medium as an isotropic elastic solid whose *only* nonlinearity is volumetric: an energy density $E(\varepsilon) = \frac{\mu}{2}J_2 + F(\det(\mathbb{1} + \varepsilon))$, with $J_1 = \text{tr } \varepsilon$ the bulk strain and J_2 the squared shear. Matching F to the deep-MOND displacement law forces F to be scale-free (a pure power law) — so nothing *inside* the constitutive law can terminate the response at high acceleration. Stronger: the high-acceleration tail required by the Saturn ephemeris bound is *exactly impossible* for any single-valued $F(\det)$ — imposing it makes the stress–strain relation double-valued (a spinodal), i.e. not a constitutive law at all. The termination must therefore come from outside the elasticity: from the entropy budget that defines the medium.

Location. Verlinde’s budget supplies it, with no free coefficient. From the published version of record (SciPost Phys. 2, 016 (2017); equation numbers below are SciPost’s; the arXiv v2 numbering (4.28)–(4.34) maps onto (28)–(34)):

- eq. (28): the entropy displaced by matter, $S_M(r) = -2\pi M c r / \hbar$ — the 2π is the Bekenstein/Unruh coefficient;
- eq. (29): the dark-energy (volume-law) entropy, $S_{DE}(r) = (r/L) A(r) / 4G\hbar$ — the $1/4G\hbar$ is the area law;
- eqs. (32)–(34): the strain–budget identification $\varepsilon_M(r) = V_M(r)/V(r) = (8\pi G/a_{0V}) \Sigma_M(r)$, with the text after eq. (42) identifying the elastic strain of the medium with this same ε_M ;
- eq. (3): the crossover criterion $\Sigma < a_{0V}/8\pi G$.

Combining (28) and (29), the ratio is exactly

$$\varepsilon_M = \frac{|S_M|}{S_{DE}} = \frac{2 G M L}{c^2 r^2} = \frac{2 g_{\text{bar}}}{a_{0V}},$$

so the budget saturates ($\varepsilon_M = 1$) at $g_{\text{bar}} = a_{0V}/2$. In the program’s units, with $a_0 = a_{0V}/Z$,

$$\boxed{y_c = \frac{g_{\text{bar}}}{a_0} \Big|_{\varepsilon_M=1} = \frac{Z}{2} = 2.894}$$

(canonical footing; 2.898 on the total-density footing, the difference pure rounding). The chain was verified against three independent renderings of the paper — the arXiv PDF, the ar5iv LaTeX, and the published SciPost version — all symbol-identical, and re-derived end-to-end in committed symbolic algebra. The one genuine $O(1)$ alternative — taking the strain to be Verlinde’s starred volume ratio with the $(d-1)/(d-2) = 3/2$ factor unabsorbed, which would give $y_c = Z/3 = 1.93$ — is closed *by the paper itself*: eq. (41) absorbs the factor and the surrounding text identifies the strain with ε_M ; section 7’s independent hydrostatic derivation agrees.

Consequences. With the minimal budget-consistent throttle — response unmodified below y_c , depleted as y_c/y above (the depletion reading; the alternative “hard cap” reading is computed dead by 4.7 orders of magnitude at Saturn, and Verlinde’s text supports depletion) — the branch passes the Saturn enclosed-mass ephemeris bound with an $18\text{--}26 \times$ margin on the strict bound, Mars and Neptune comfortably, with the baryonic Tully–Fisher zero-point untouched to machine precision (all outermost SPARC points lie below y_c) and the deep-regime 0.982 amplitude preserved exactly.

3. Result 2: the medium’s shear sector cannot be linearized general relativity

An attractive route to suppressing the medium’s Cassini quadrupole would be for its shear sector to *be* linearized GR — shear waves = gravitational waves, Planck-stiff, so the galactic-strain anharmonicity is negligible by hierarchy. This is refuted on the program’s own normalizations, with an exact closed form. Linearized GR, read as an elastic medium, pins only the *ratio* $\mu/\rho = c^2$ (the metric perturbation is dimensionless strain; GR supplies no intrinsic modulus), so with the medium’s inertial density ρ_Λ : $\mu_{GW} = \rho_\Lambda c^2 = 5.25 \times 10^{-10}$ Pa. But the deep-MOND energy match — using the derived compatibility lemma that *no decaying radial displacement field is shear-free* (for $u \sim r^{-n}$, $J_2/J_1^2 = \frac{2}{3}(n+1)^2/(2-n)^2$; the deep-MOND profile has $J_2 = J_1^2/6$ exactly) — pins the medium’s *total* deep stiffness at $K_{\text{eff}} = a_0^2/16\pi G$, capping the shear modulus at least $33\text{--}134 \times$ (convention-dependent) below μ_{GW} :

$$\frac{\rho_\Lambda c^2}{K_{\text{eff}}} = 6Z^2 = 201.1 \quad (\text{exact}).$$

The fork is closed in both directions: insisting on GR-stiff shear collapses the deep lensing amplitude from 0.982 to 0.119 (excluded by the rotation-curve/lensing data the branch exists to fit); keeping the amplitude forces the medium’s shear waves to $v \lesssim 0.17c$ — material sound in a source medium, *not* gravitational waves. (Metric gravitational waves still propagate at c on the one shared metric; GW170817 is untouched. What dies is only the identification, and with it the free suppression of the shear anharmonicity — which therefore remains a bounded, posited quantity, straddling the Cassini requirement; see §5.)

4. The observable fingerprint — located, not detected

The cutoff’s signature in the radial-acceleration relation is fully characterized: *exactly zero* deviation from the uncut law below $y_c = 2.894$ (so the BTFR zero-point and the $y \lesssim 3$ RAR curvature are untouched), a kink at y_c , and a peak deviation of 0.017 dex at $y \approx 6$, decaying beyond. Against the full SPARC point-level sample (175 galaxies, 3,391 points, mass-to-light refit per the program’s

standard protocol, both footings): the post-refit separation between the throttled and uncut laws on the combined $y = 4\text{--}30$ statistic is 0.0148 dex against a present error of 0.025–0.029 dex – 0.5–0.6 σ , **indistinguishable**. Moreover the discriminating window is two-thirds bulge-dominated, and the bulge mass-to-light systematic (freeing the bulge Υ from the conventional $1.4\Upsilon_{\text{disk}}$ lock shifts the statistic by 0.037 dex) is $2.5 \times$ the entire signal; a formal fit preference for the throttled law that appears under the locked convention is monotone in generic suppression – it does not select $y_c = Z/2$ – and is therefore an artifact, not a detection. No data preference in either direction is claimed. The decisive test is specified: $\sigma \lesssim 0.005$ dex on the high- y statistic, requiring roughly $25\text{--}34 \times$ the current high- y sample *and* external stellar-population bulge mass-to-light priors at the same level – i.e., a bulge-rich rotation-curve sample of BIG-SPARC class, not further reanalysis of existing data.

5. The honest ledger

What is **derived** here: the necessity of the response cutoff (scale-free constitutive law + the single-valued- F impossibility theorem); its location $y_c = Z/2$ from Verlinde’s own published coefficients, with the $Z/3$ alternative excluded by his own eq. (41); the Saturn/BTFR/deep-amplitude consequences of the minimal throttle; and the exact $6Z^2$ refusal of the GR-stiff-shear identification. What is **transplanted** (named, not derived): the pure- Λ horizon reading ($a_{0V} = cH_\Lambda$; Verlinde’s own horizon is cH_0 , eq. (2)) – on the canonical footing the exactness of $Z/2$ is definitional given $a_0 := cH_\Lambda/Z$, and the paper-pinned physics is precisely the coefficient 2. What is the **program’s own construction** (not Verlinde’s claim): the throttle form $T = \min(1, y_c/y)$ – only the location and the necessity of termination are his – and the depletion (versus cap) reading, supported by his text and by the cap being computed dead. What remains **posited**: the medium’s shear-anharmonicity fraction (bounded by an adversarially verified band that *straddles* the Cassini quadrupole requirement – the branch is neither cleared nor killed there), and the absence of any covariant action for the medium (the branch’s structural debt). What remains **open program-wide**: the Branch A/Branch B identity choice itself; the MOND sign, a_0 ’s value, and Z remain posited inputs; the Standard Model is untouched; and nothing here re-asserts any previously retracted claim. **This note locates a signature; it does not detect one, and it does not adopt the branch.**

Reproducibility

All results are backed by committed, runnable scripts (repository `real_research/reviews/shear_linearity_2026/`): the coefficient pin and end-to-end re-derivation (`p6_pin_verlinde_coefficients.py`, `p6_pin_adversarial_recheck.py` – three-rendering source verification), the SPARC fingerprint confrontation (`p6_throttle_fingerprint_sparc.py`), the shear-linearity campaign (`lane1_theorem.py`, `lane2_gates.py`, `lane3_tail_falsifiers.py`), and the underlying gate machinery (`./branchB_q2_gate_2026/`, including the corrected Milgrom/Desmond quadrupole kernel validated against Desmond et al.’s published anchors). Companion records of the program: Zenodo concept DOI 10.5281/zenodo.21253644.

References

1. E. Verlinde, *Emergent Gravity and the Dark Universe*, SciPost Phys. **2**, 016 (2017); arXiv:1611.02269. (Equation numbers cited are the SciPost version’s.)
2. M. Milgrom, ApJ **270**, 365 (1983); Phys. Lett. A **253**, 273 (1999).
3. S. Deser & O. Levin, Class. Quantum Grav. **14**, L163 (1997).

4. F. Lelli, S. McGaugh & J. Schombert, AJ **152**, 157 (2016); MNRAS Lett. **468**, L68 (2017) (the emergent-gravity RAR test).
5. S. Boran, S. Desai, E. Kahya & R. Woodard, Phys. Rev. D **97**, 041501 (2018) (GW170817 vs dark-matter emulators).
6. H. Desmond, A. Hees & B. Famaey, MNRAS **530**, 1781 (2024); R. Park et al., arXiv:2602.17884 (2026) (the solar-system quadrupole bound).
7. M. Brouwer et al., A&A **650**, A113 (2021); T. Mistele & S. McGaugh, arXiv:2310.15248 (2024) (lensing tracks dynamics).
8. Pitjev & Pitjeva, Astron. Lett. **39**, 141 (2013) (ephemeris bound on unmodeled mass).

Both a_0 footings carried throughout. Each claim labeled derived / transplanted / constructed / posited. This note prices a branch of the program; it does not adopt it, does not claim a detection, and makes no completeness or theory-of-everything claim.